POWER_CLOCK_RESET

File: power-clock.kicad_sch

CPU

File: cpu.kicad_sch

ROM

File: rom.kicad_sch

RAM

File: ram.kicad_sch

GLUE

File: glue.kicad_sch

COMPACT_FLASH

File: compact-flash.kicad_sch

IO

File: io.kicad_sch

SERIAL

File: serial.kicad_sch

RTC

File: rtc.kicad_sch

PORTS

File: ports.kicad_sch

PANEL

File: panel.kicad_sch

DEBUG_HEADER

File: debug-header.kicad_sch

ENGINE

File: engine.kicad_sch

PIXEL

File: pixel.kicad_sch

COMPOSITOR

File: compositor.kicad_sch

VGA_OUT

File: vga.kicad_sch

AUDIO

File: audio.kicad_sch

AUDIO_FIFO

File: audio-fifo.kicad_sch

AUDIO_DAC

File: audio-dac.kicad_sch

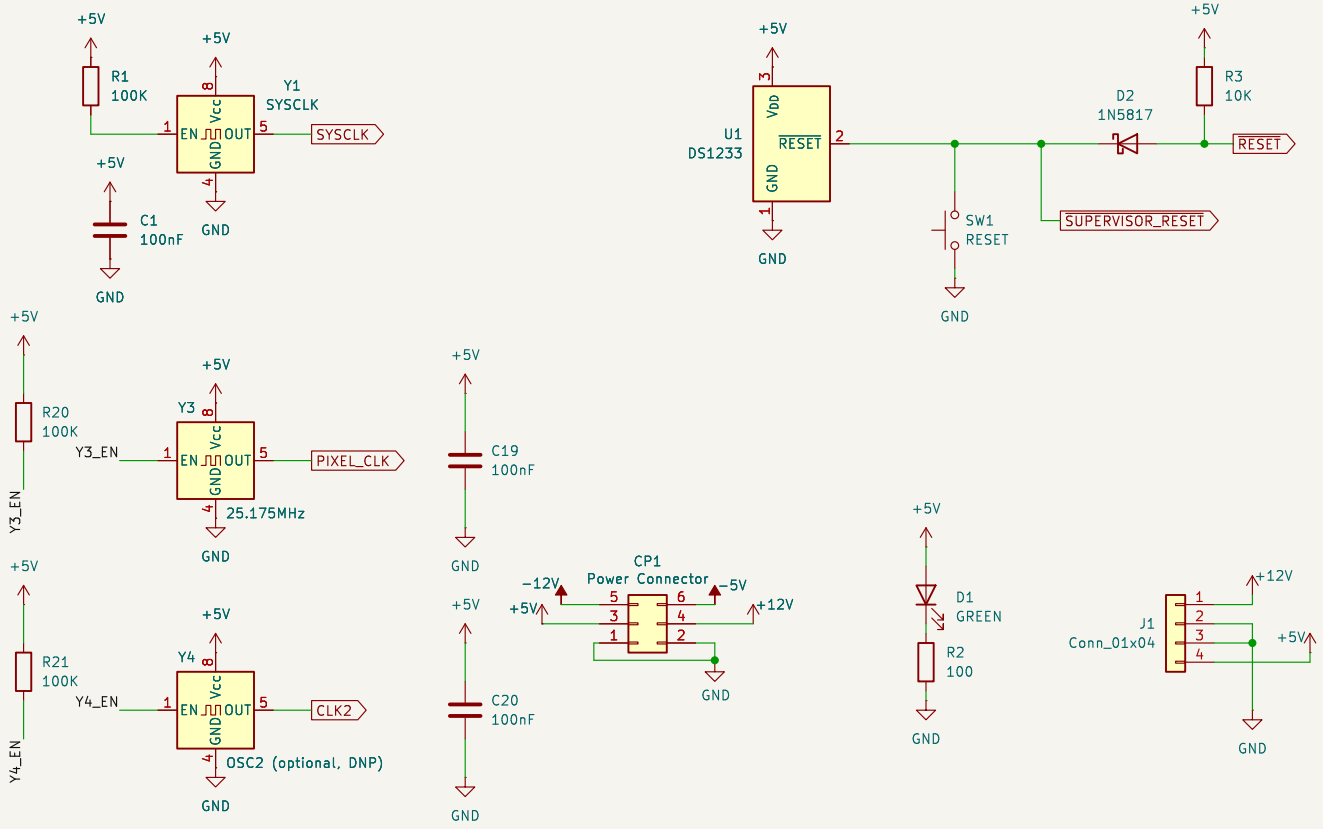
AUDIO_AMP

File: audio-amp.kicad_sch

Sheet: /
File: board.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 1/21



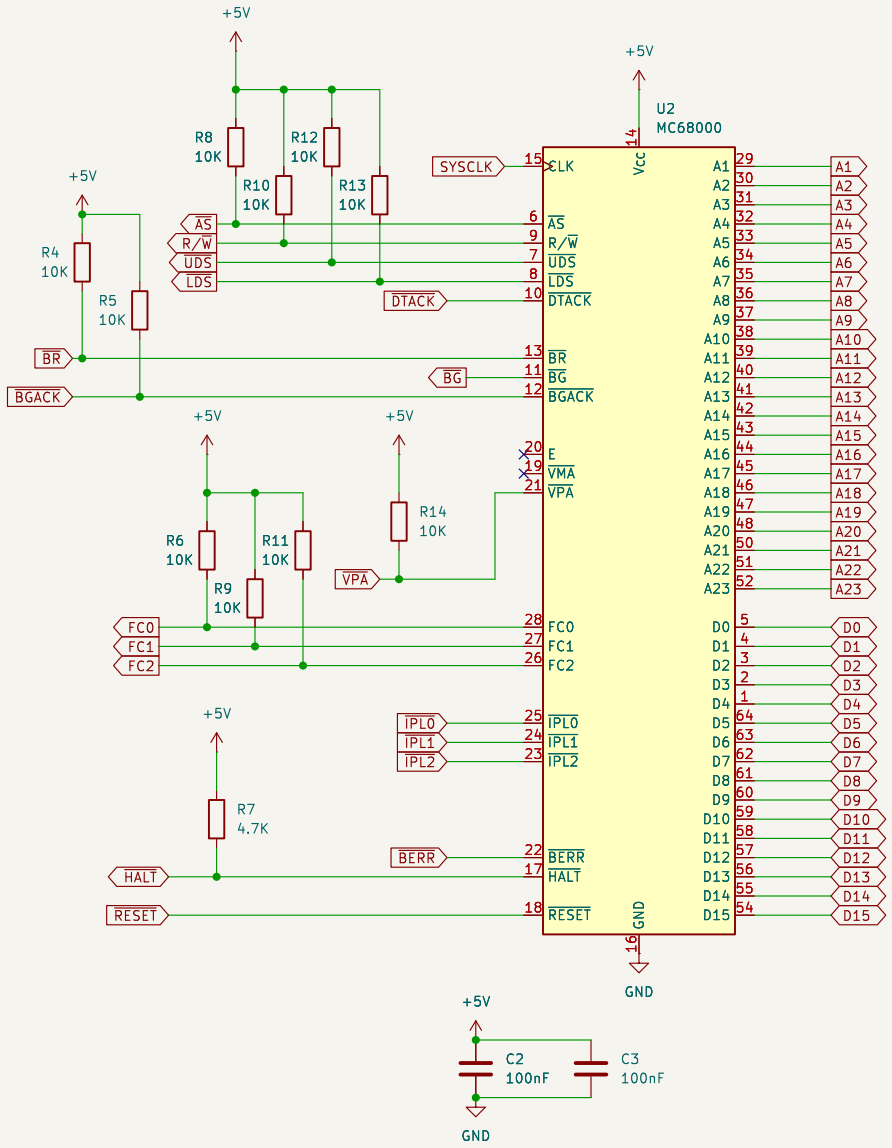
Datasheet oscillators are NC on pin 1 but some of my oscillators in hand are ENB on pin 1, so pull-up as a hedge. Y4 is the optional second oscillator on PIXEL GCLK2 (pin 2), unstuffed by default.

H1 MountingHole GND
H2 MountingHole GND
H3 MountingHole GND
H4 MountingHole GND
H5 MountingHole GND

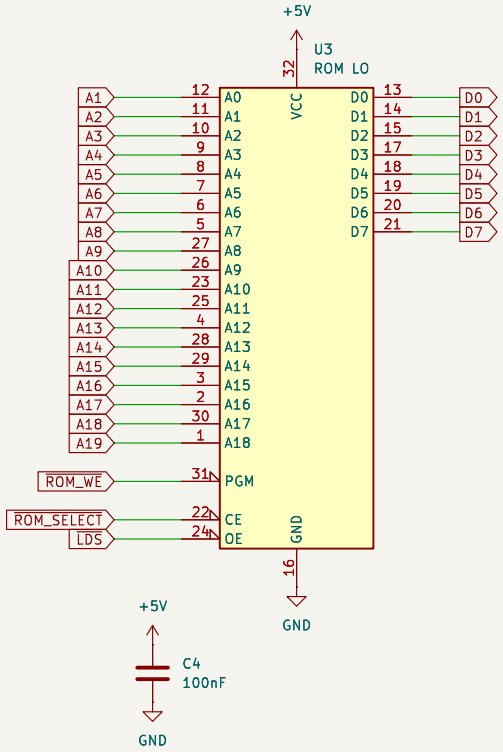
Sheet: /POWER_CLOCK_RESET/
File: power-clock.kicad_sch

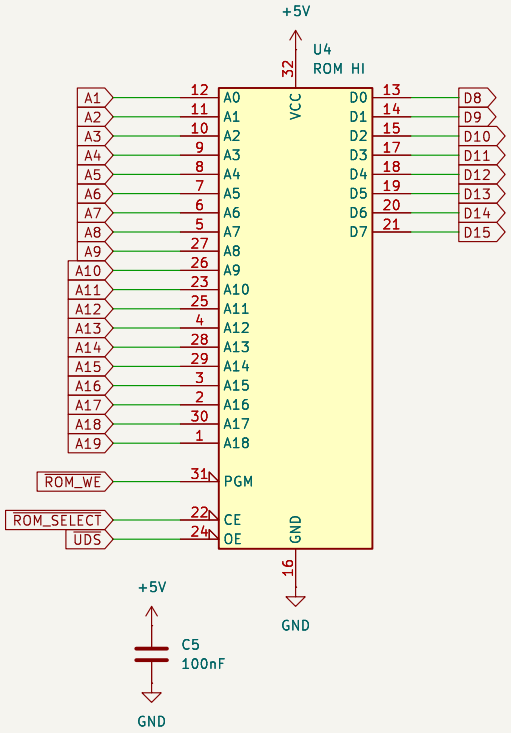
Title:

Size: USLetter | Date: | **Rev:**
KiCad E.D.A. 9.0.7 | Id: 2/21



Sheet: /CPU/	
File: cpu.kicad_sch	
Title:	
Size: USLetter	Date:
KiCad E.D.A. 9.0.7	Id: 3/21



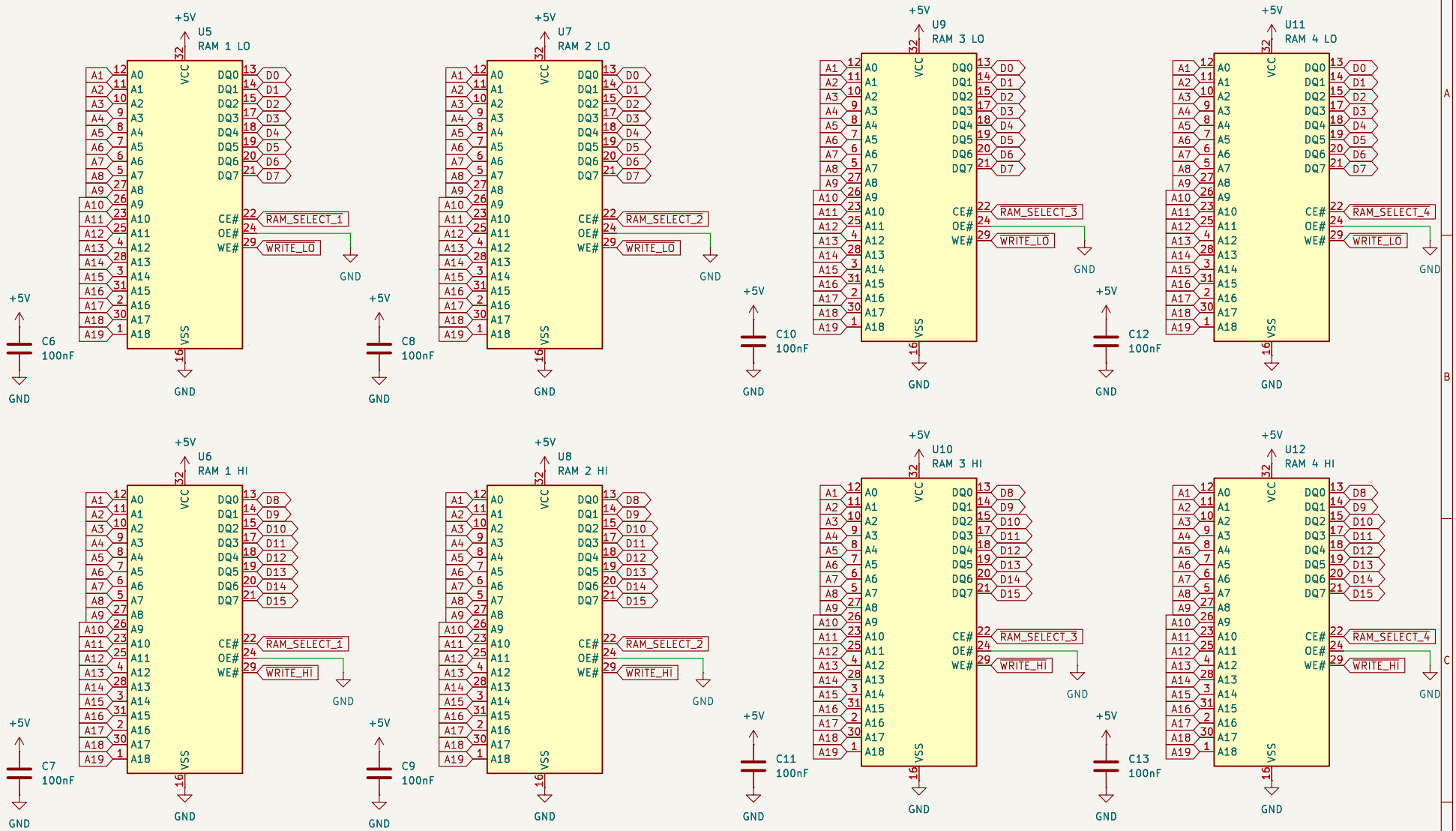


Sheet: /ROM/
File: rom.kicad_sch

Title:

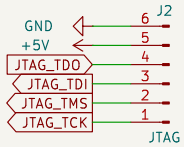
Size: USLetter | Date: | **Rev:**

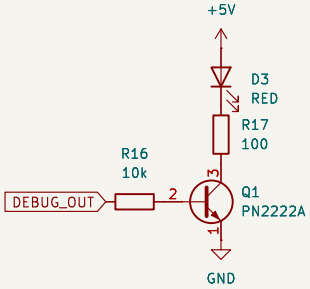
KiCad E.D.A. 9.0.7 | Id: 4/21

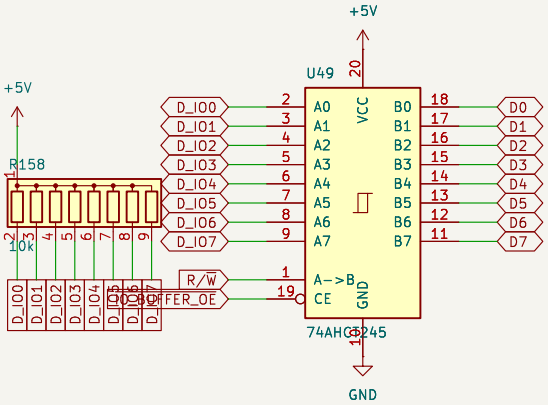


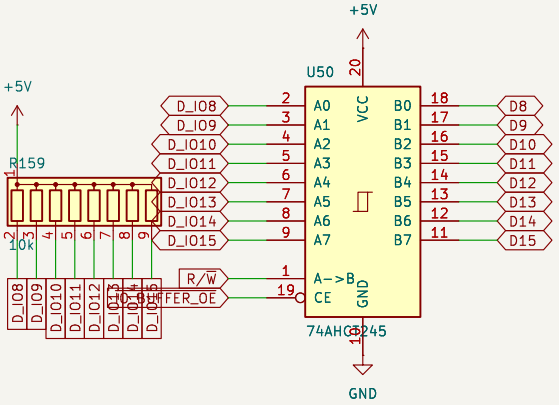
Sheet: /RAM/	
File: ram.kicad_sch	
Title:	
Size: USLetter	Date:
KiCad E.D.A. 9.0.7	Id: 5/21

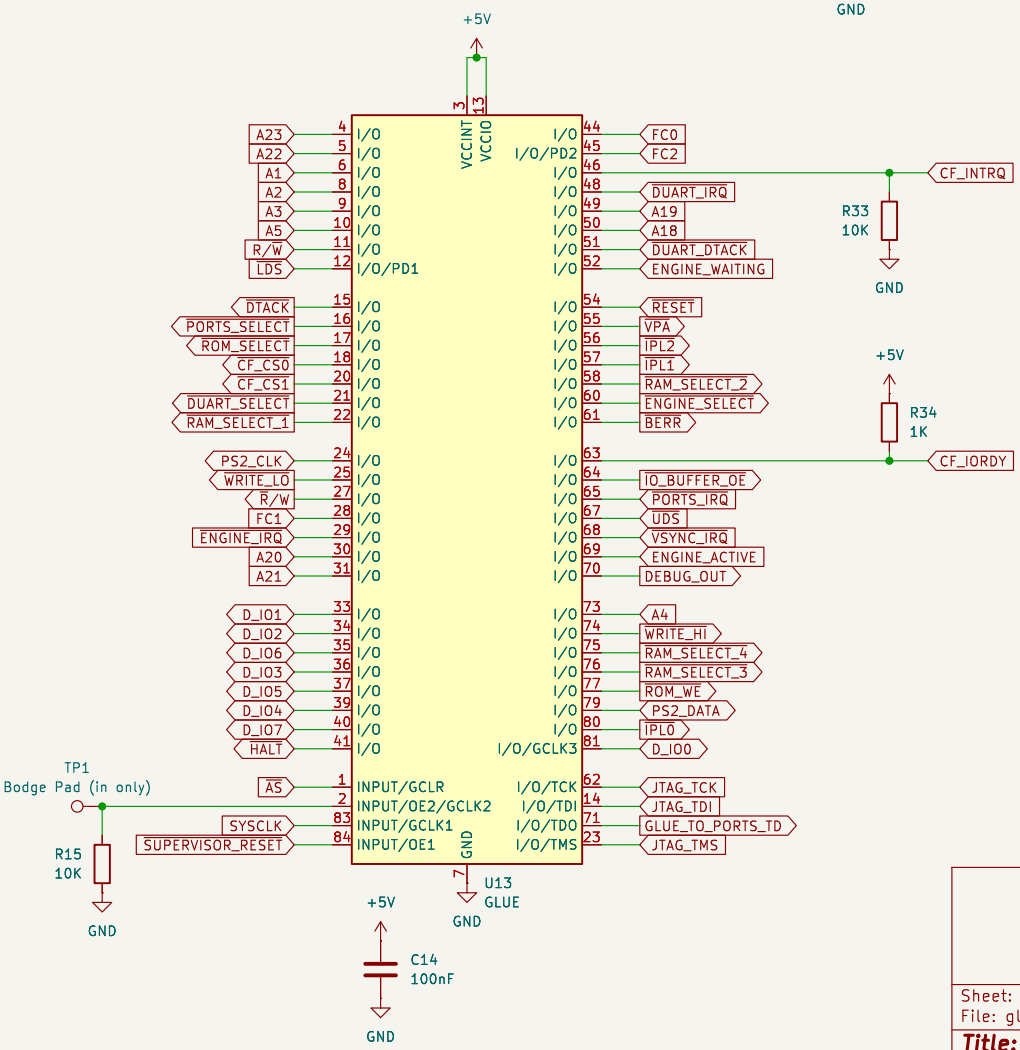
J2 JTAG 1x6: TCK TMS TDI TDO +5V GND.
An Adafruit FT232H breakout drives it
from D0 TCK, D1 TDI, D2 TDO, D3 TMS + GND.

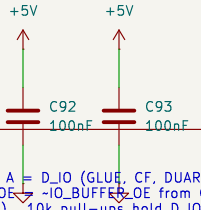










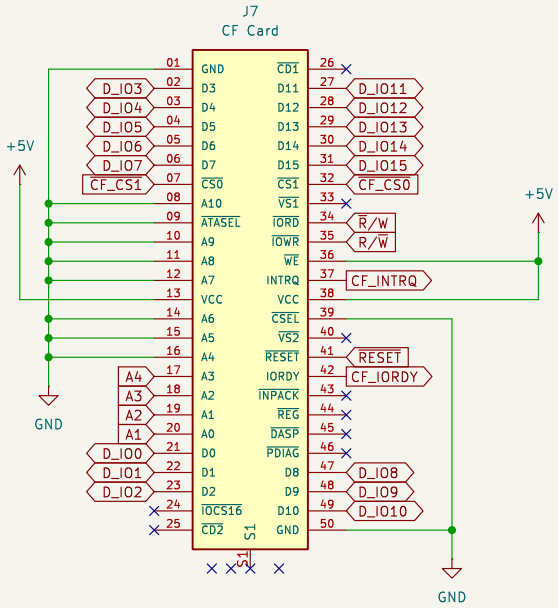


IO data buffers: 74AHCT245 pair, A = D_IO (GLUE, CF, DUART, PORTS), B = main bus D; DIR = R/~W, /OE = ~IO_BUFFER_OE from GLUE (low for any cycle in 0xF00000–0xFFFFFF). 10k pull-ups hold D_IO when the buffers are disabled (griffin.yml "IO data buffers").

Sheet: /GLUE/
File: glue.kicad_sch

Title:

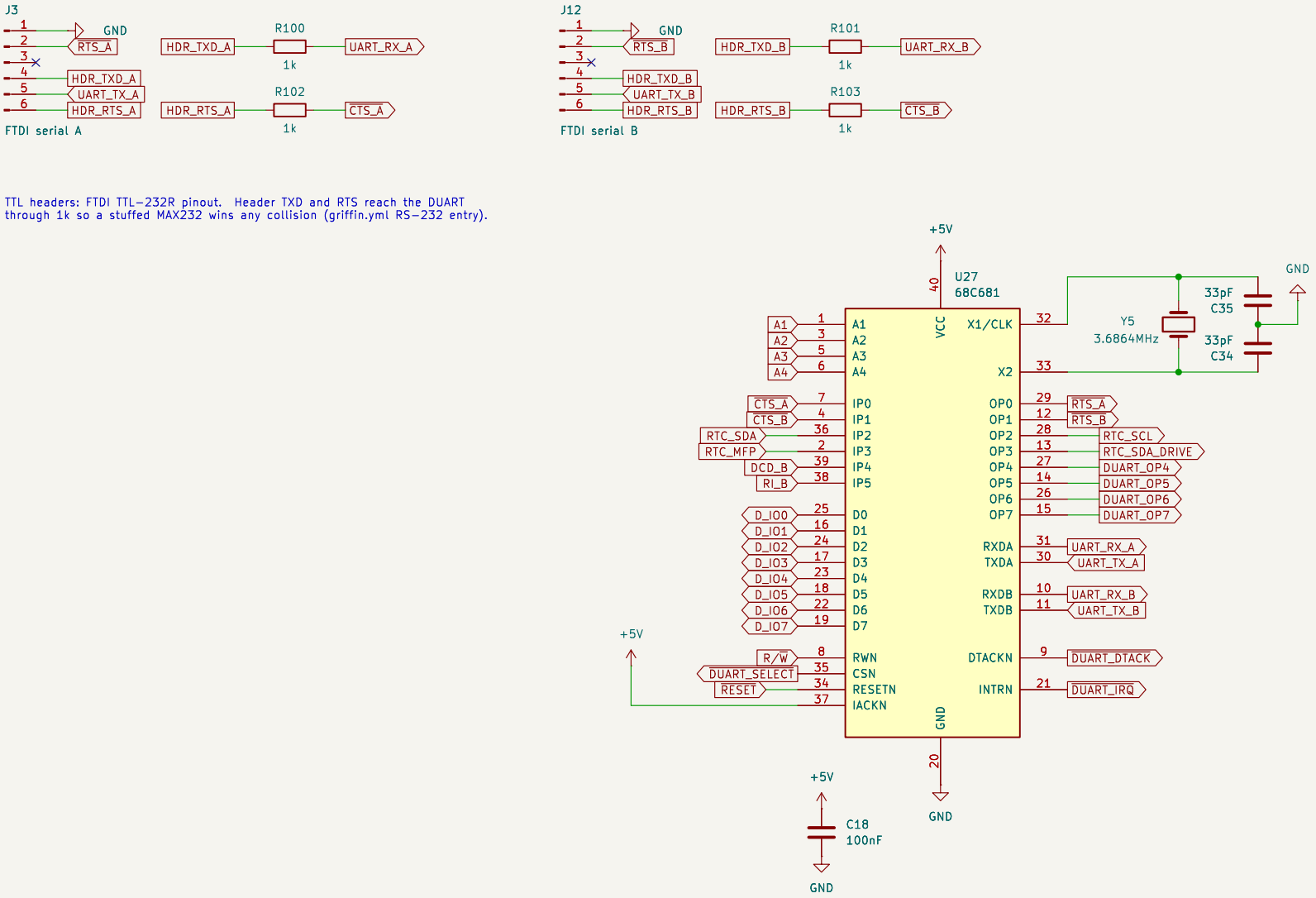
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 6/21



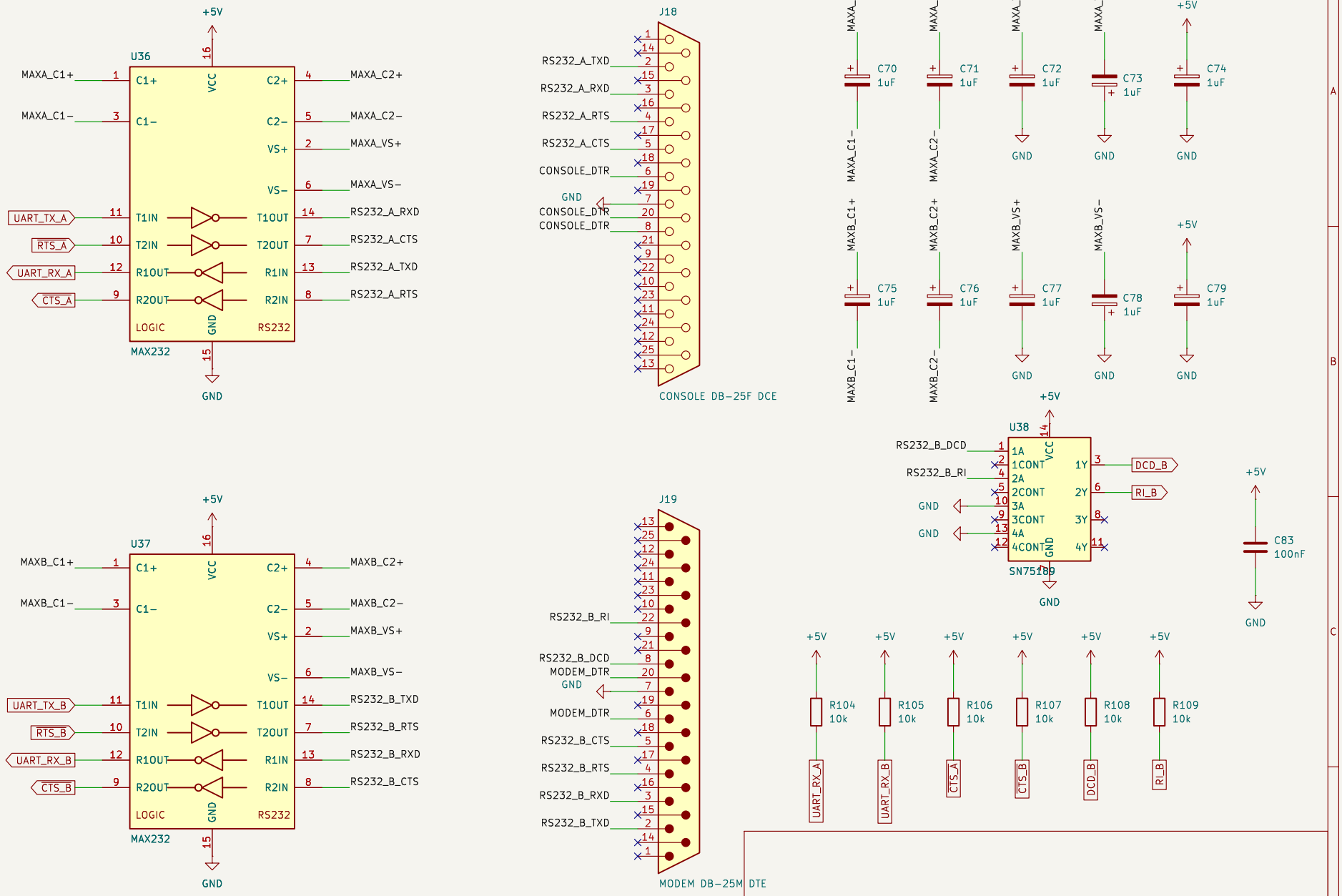
Sheet: /COMPACT_FLASH/
File: compact-flash.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 7/21



Sheet: /IO/	
File: io.kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 9.0.7	Id: 8/21

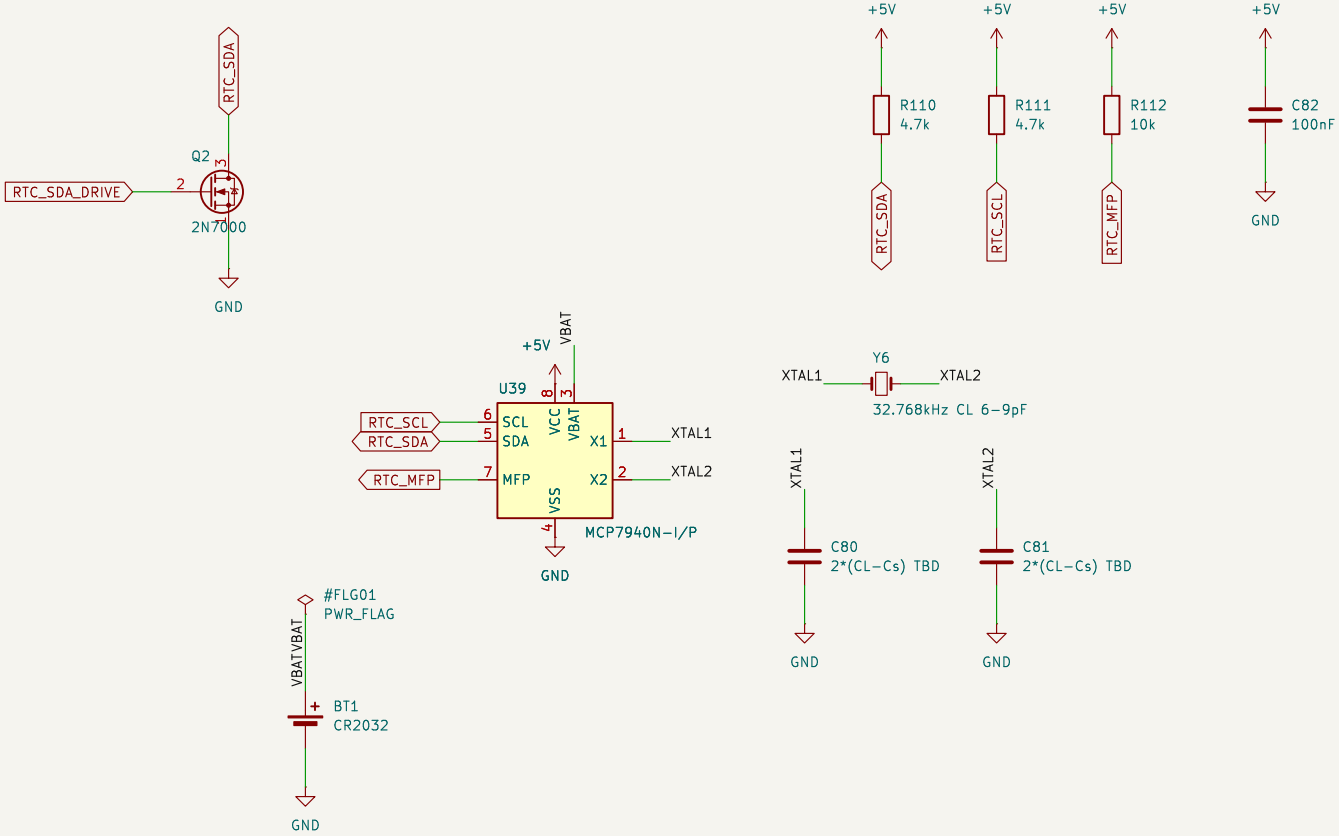


RS−232 level stages (griffin.yml interfaces, RS−232 level stage).
Ch A console: DB−25 female wired DCE; terminal TXD (2) and RTS (4) are received, TXDA and ~RTS_A
drive pins 3 (RXD) and 5 (CTS); DSR (6) and DCD (8) strapped to the terminal DTR (20).
Ch B modem: DB−25 male wired DTE; TXDB −> 2, 3 −> RXDB, ~RTS_B −> 4, 5 −> ~CTS_B,
DCD (8) and RI (22) through the 75189 to IP4/IP5 (unused 75189 inputs grounded); DTR (20) strapped to DSR (6).
Local net names follow the DB−25 pin signal (DTE−referenced). Pin 7 signal ground.
The TTL headers share these nets (io sheet, 1k series); 10k pull−ups hold RXD, ~CTS, DCD_B and RI_B
idle when the level stages are unstuffed. No isolation jumpers.

Sheet: /SERIAL/
File: serial.kicad_sch

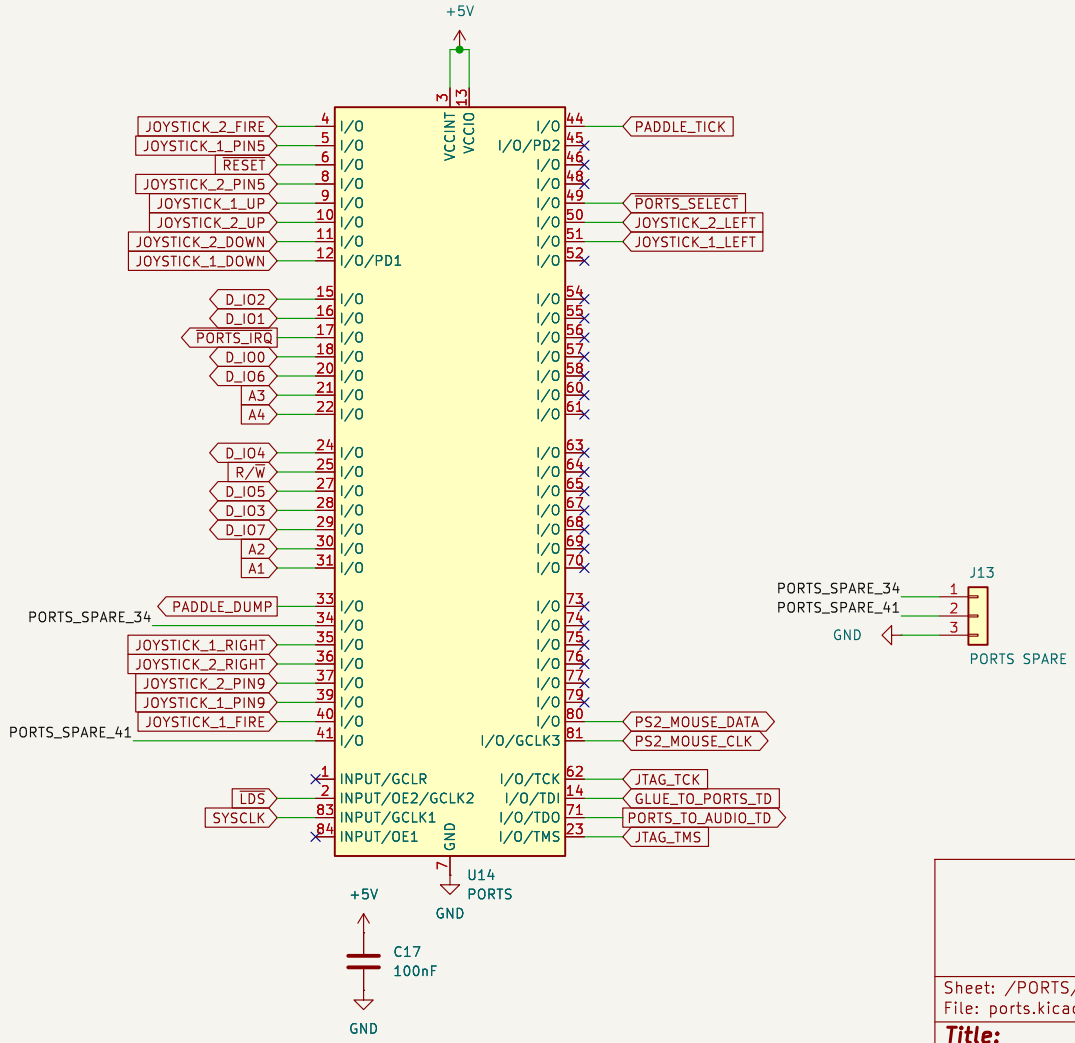
Title:

Size: USLetter	Date:	Rev:
KiCad E.D.A. 9.0.7		Id: 9/21

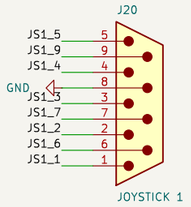


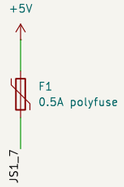
SCL = DUART OP2 (push-pull: the MCP7940N is the only I2C device and does not clock-stretch);
SDA driven low through Q2 by OP3 (OPR_SET drives the pin LOW, FET inverts again), read back on IP2;
MFP -> IP3, firmware programs the 1 Hz square wave. Pull-up values are assumptions, not in griffin.yml.

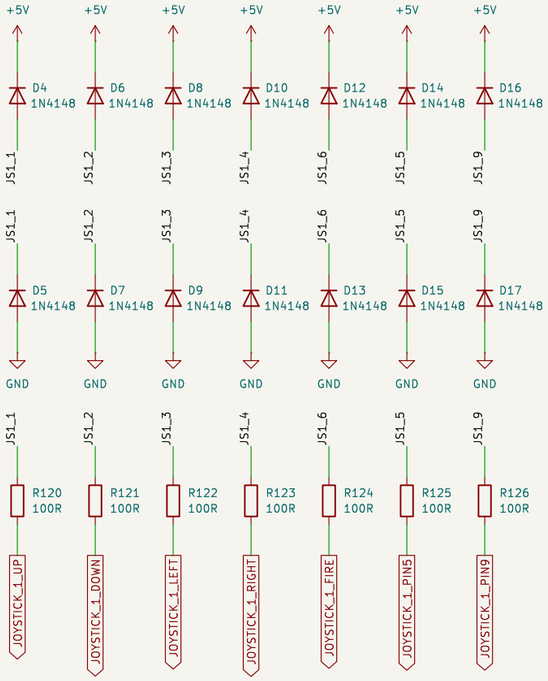
Sheet: /RTC/		
File: rtc.kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 10/21

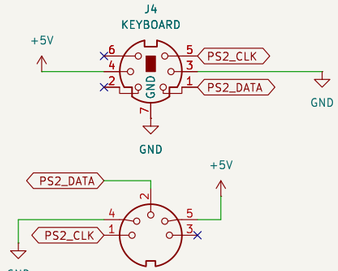


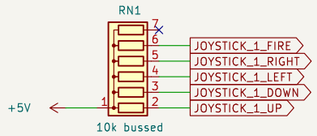
Sheet: /PORTS/	
File: ports.kicad_sch	
Title:	
Size: USLetter	Date:
KiCad E.D.A. 9.0.7	Rev: Id: 11/21

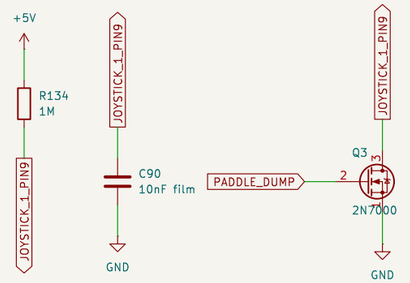






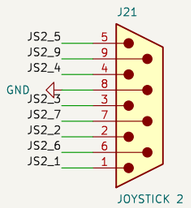


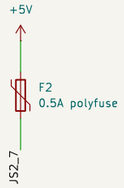


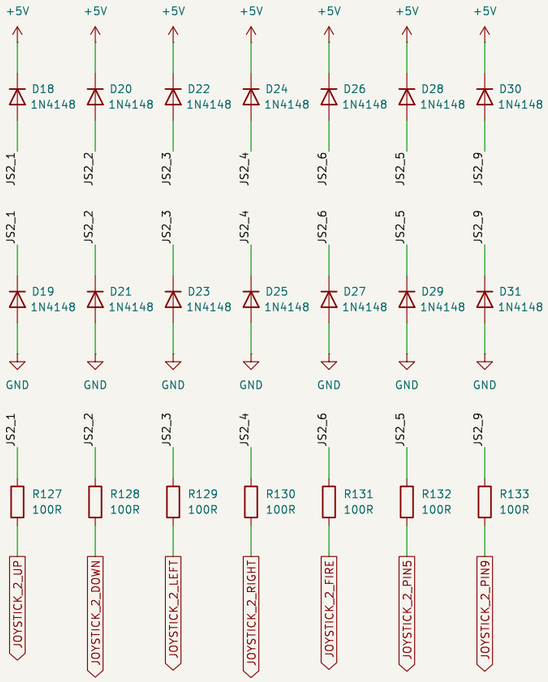


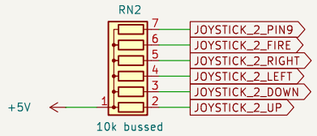
Atari DE-9: 1 up, 2 down, 3 left, 4 right, 5/9 paddle pots (SMS button 2 on 9), 6 fire, 7 +5V (polyfuse), 8 GND. Port 1 hosts the paddles: 1M pull-ups on 5/9 (the pot parallels the pull-up), 10 nF ramp caps, 2N7000 dump FETs on PADDLE_DUMP; DNP the caps/FETs for a sticks-only build.

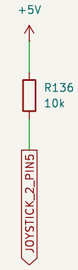
ESD: 100R series + 1N4148 clamps to +5V/GND on every switch line, connector side. Values 100R, 1N4148, 10k, 1M, 10 nF and 0.5 A are assumptions; griffin.md gives only the topology.







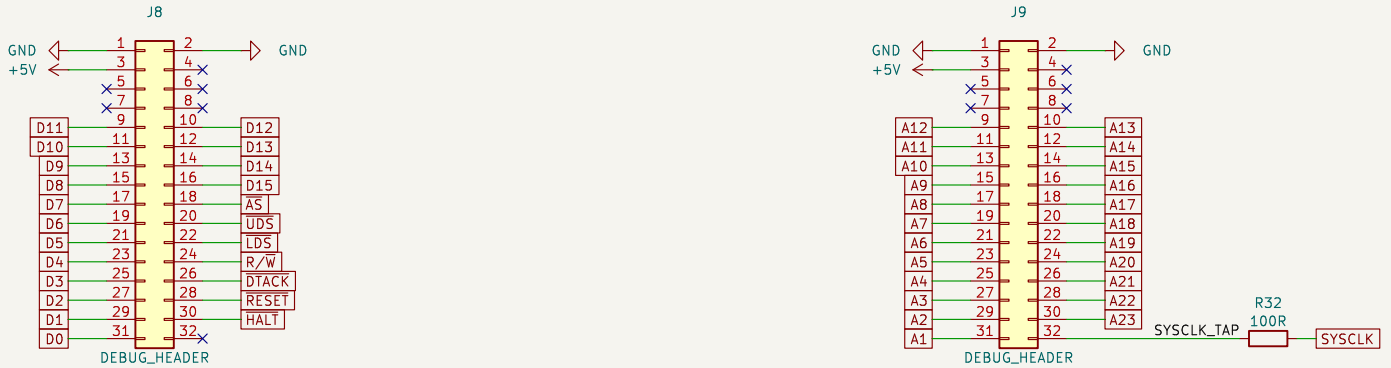




Sheet: /PANEL/
File: panel.kicad_sch

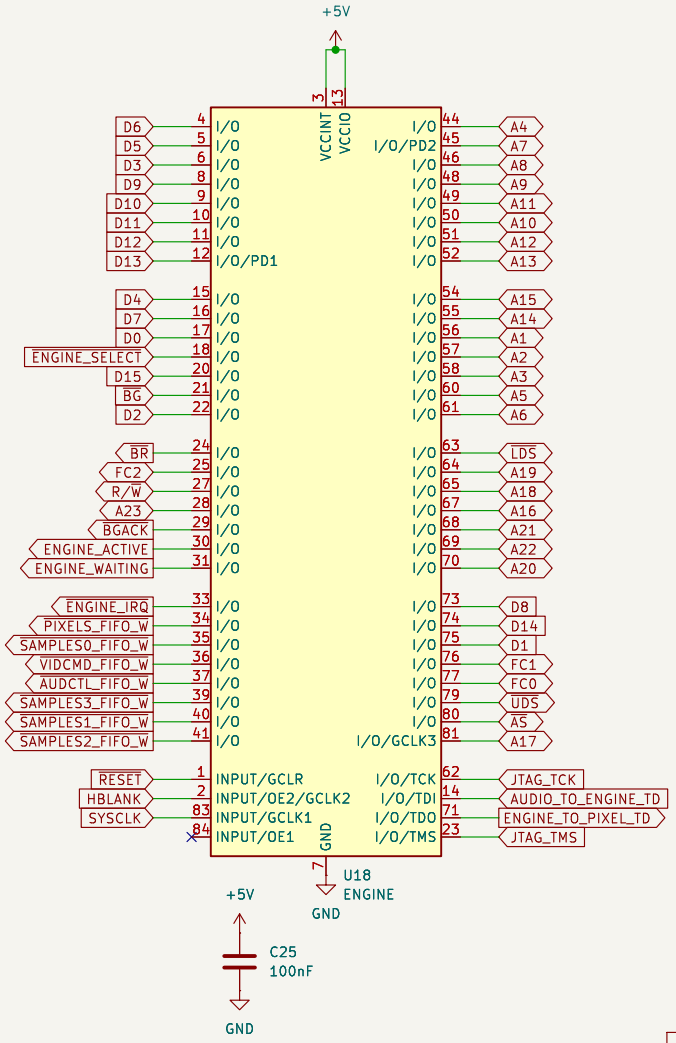
Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 12/21

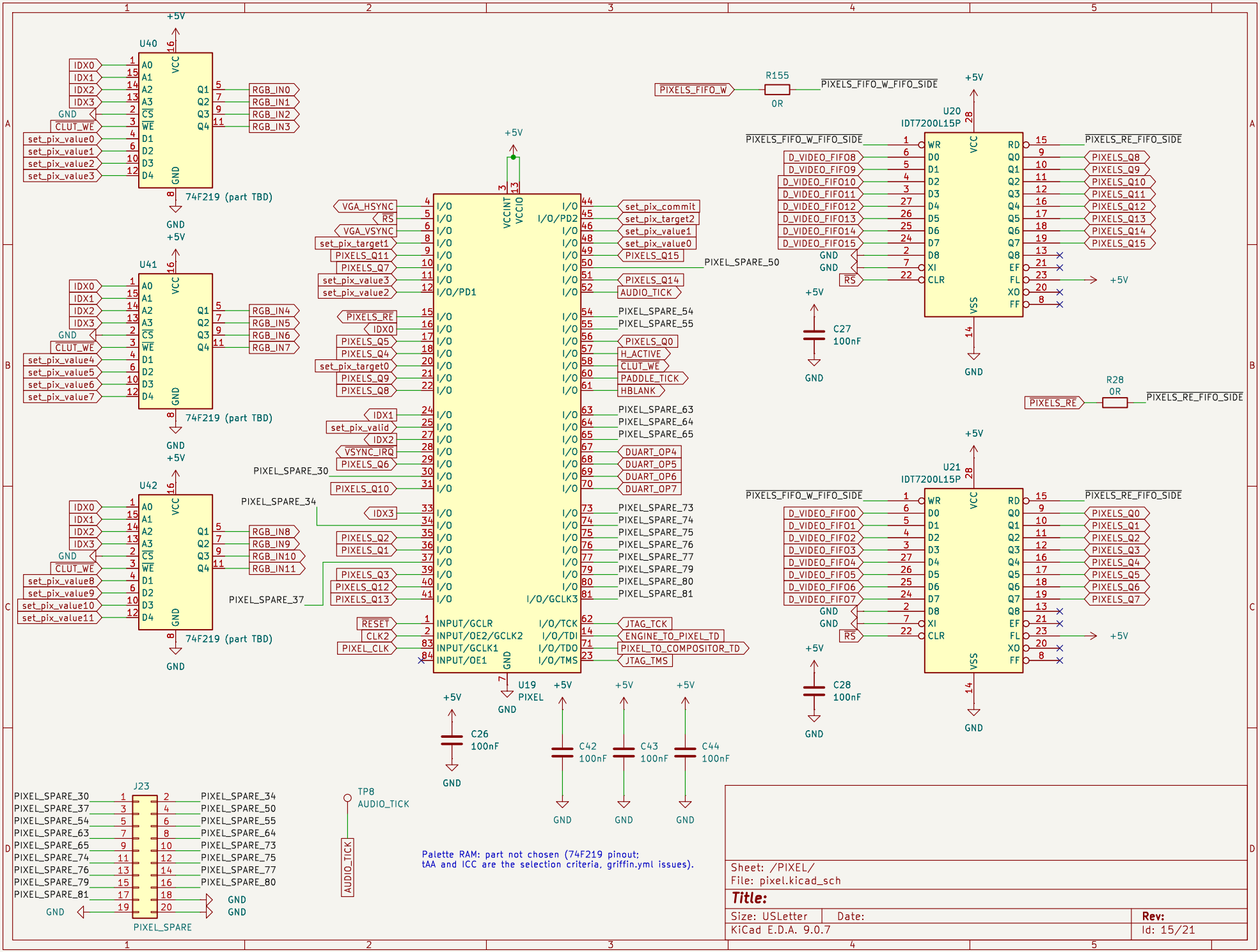


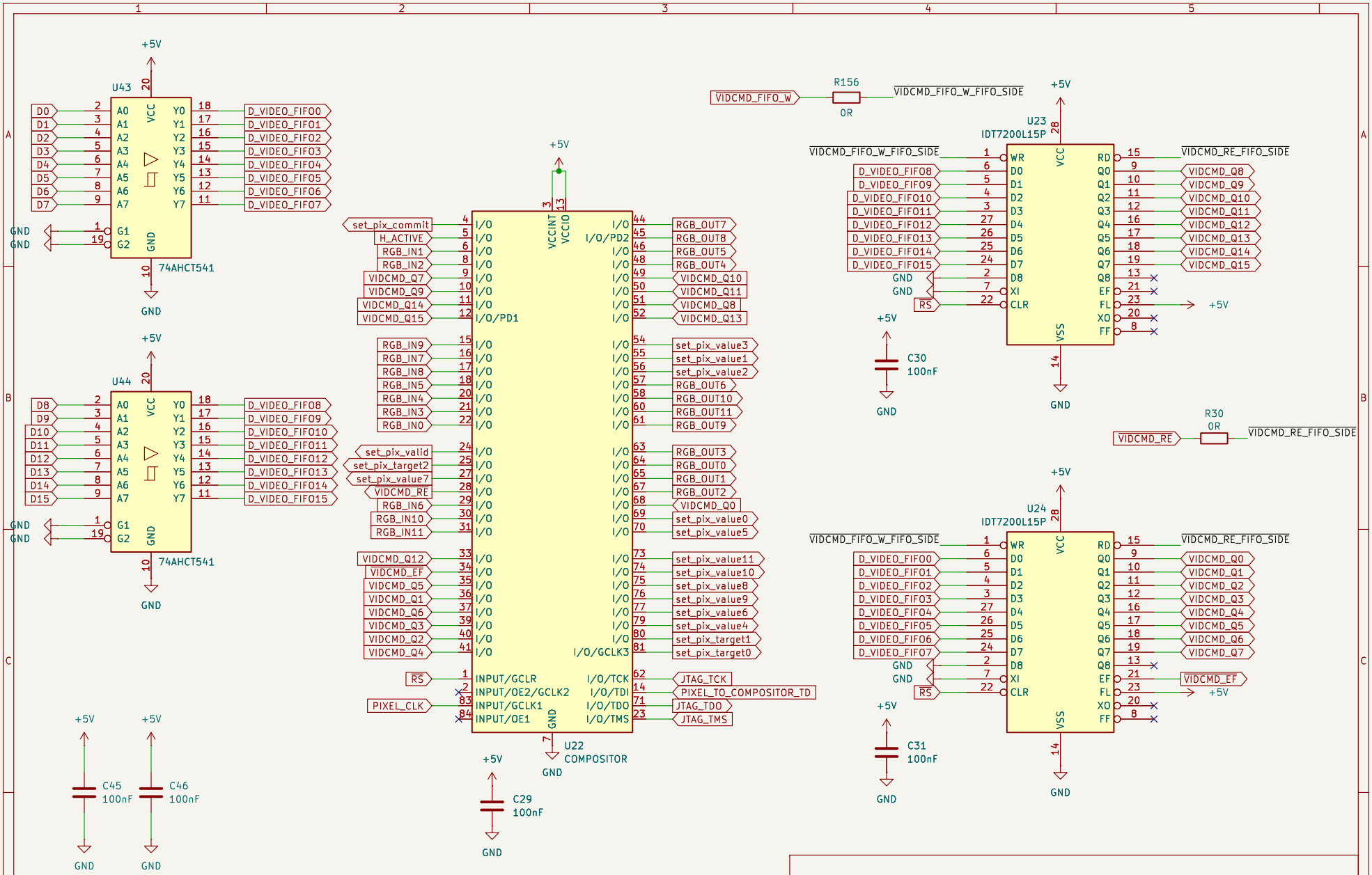
Debug headers mirror the gusmanb LogicAnalyzer V2 input connector (2x16 shrouded, 32-way IDC ribbon straight through).
DINk = analyzer channel k. Pins 5-8 are the analyzer's own 5 V / 3V3 and stay open; pin 3 feeds its EXT_VREF from +5V.
J8: ch 1-16 D0-D15, 17-23 ~AS ~UDS ~LDS R/~W ~DTACK ~RESET ~HALT, 24 spare. J9: ch 1-23 A1-A23, 24 SYSCLK via R32.

Sheet: /DEBUG_HEADER/		
File: debug-header.kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 13/21



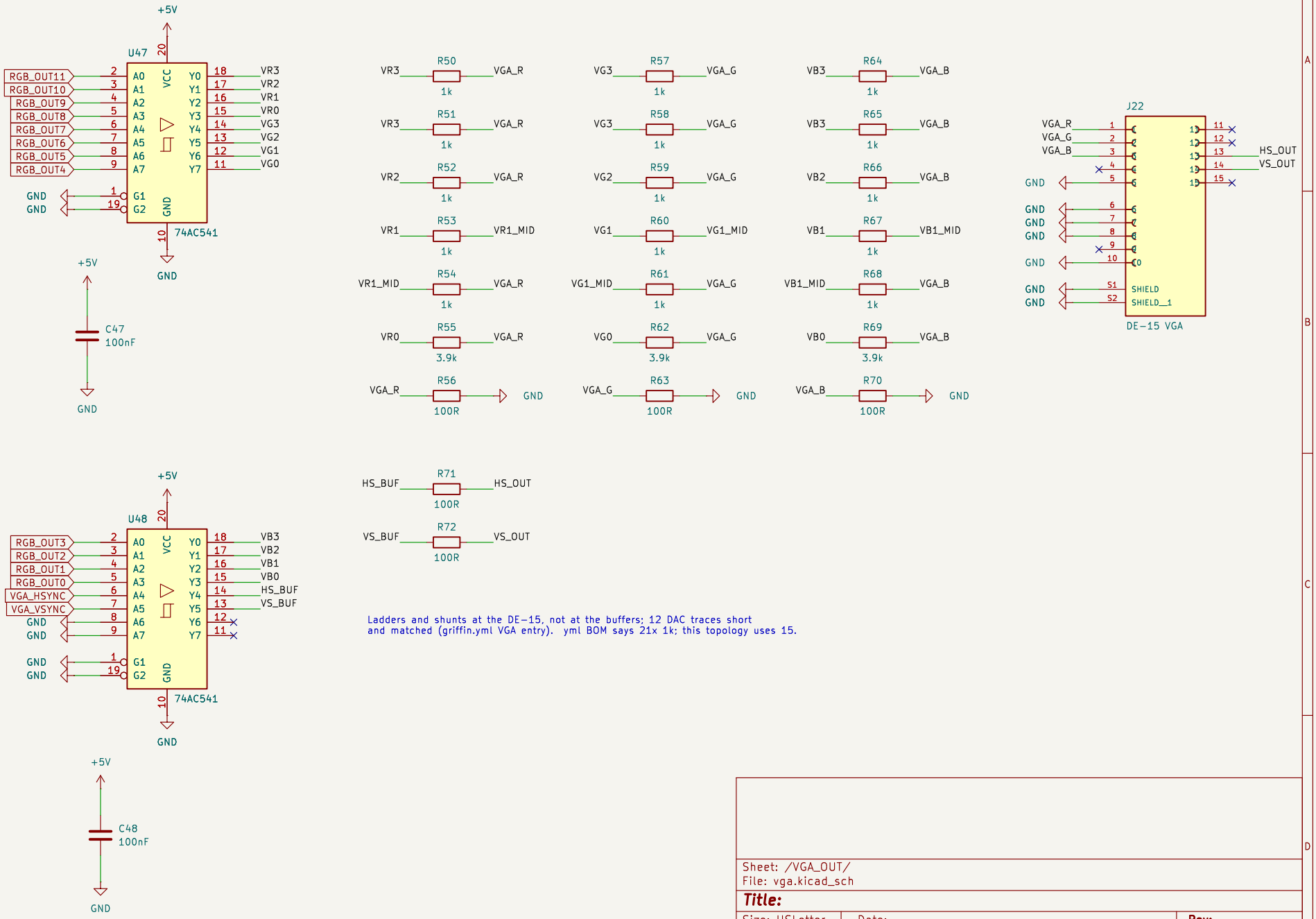
Sheet: /ENGINE/	
File: engine.kicad_sch	
Title:	
Size: USLetter	Date:
KiCad E.D.A. 9.0.7	Id: 14/21

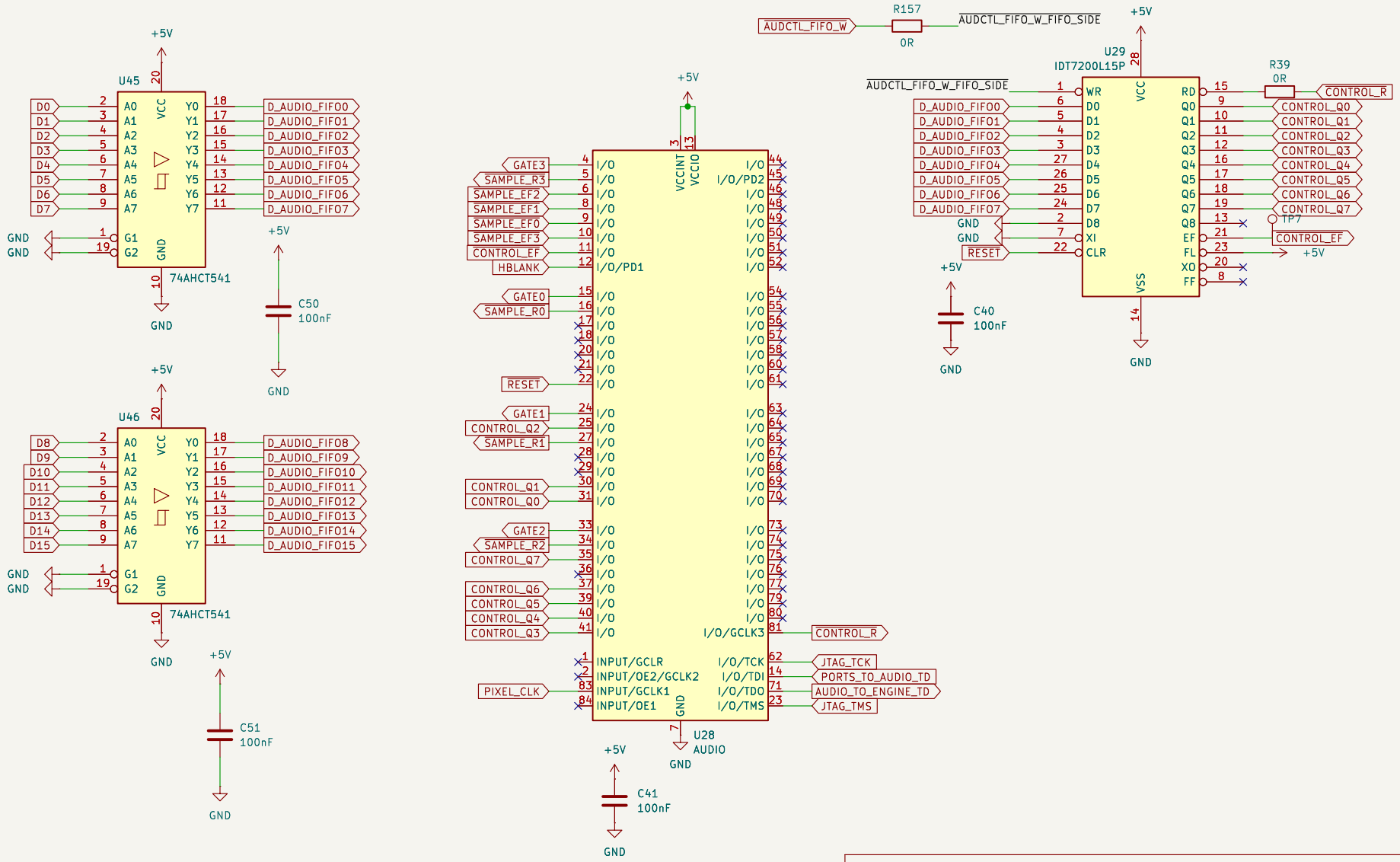

U40
74F219 (part TBD)
U41
74F219 (part TBD)
U42
74F219 (part TBD)
U19
PIXEL
U20
IDT7200L15P
U21
IDT7200L15P
R155
0R
R28
0R
C26
100nF
C27
100nF
C28
100nF
C42
100nF
C43
100nF
C44
100nF
J23
PIXEL_SPARE
TP8
AUDIO_TICK
Palette RAM: part not chosen (74F219 pinout;
tAA and ICC are the selection criteria, griffin.yml issues).
Sheet: /PIXEL/
File: pixel.kicad_sch
Title:
Size: USLetter
Date:
Rev:
KiCad E.D.A. 9.0.7
Id: 15/21

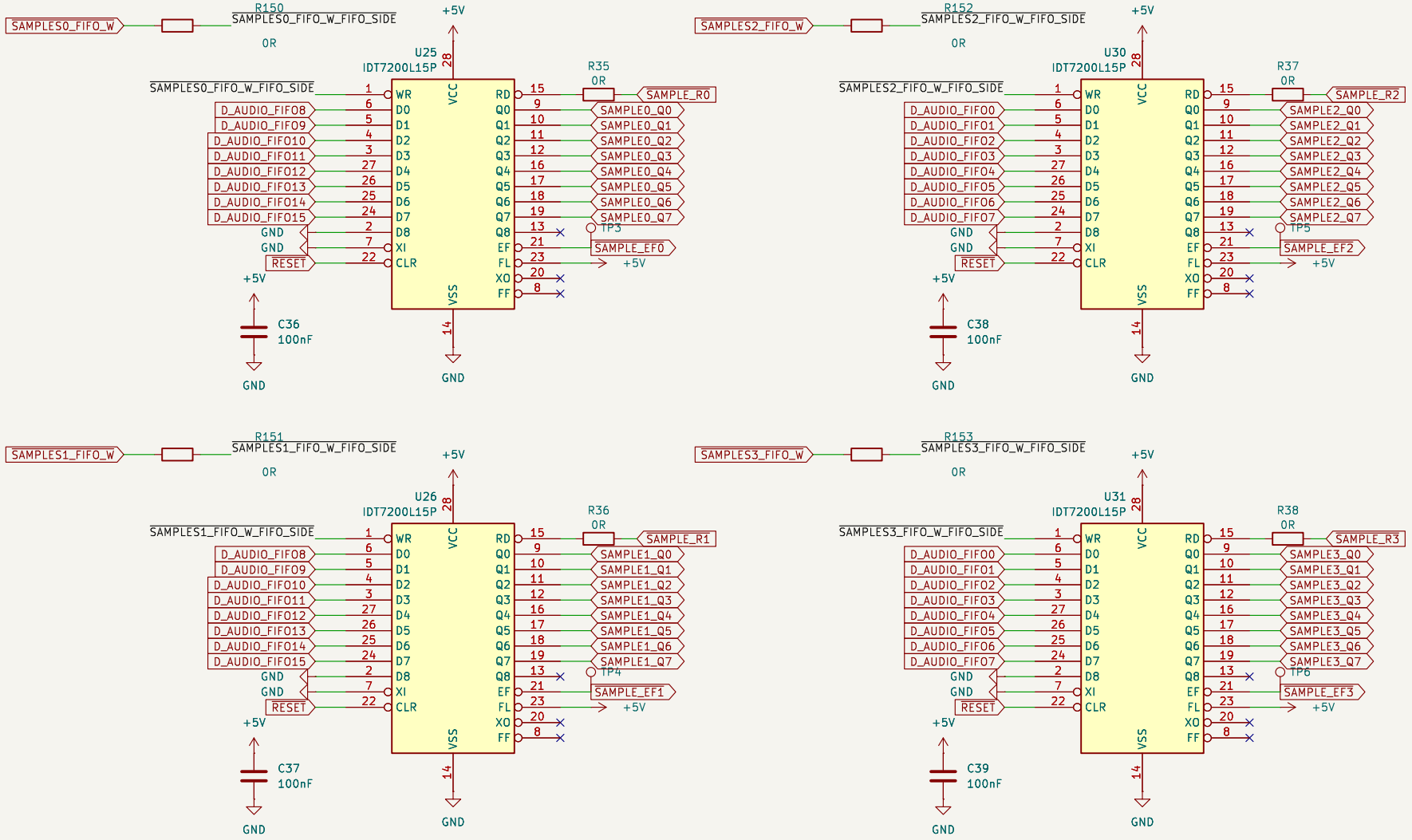


Video FIFO write buffers: AHCT, not HCT (the buffer delay comes out of data setup at the 7200 /W edge); write-data path only, /W /RS /R /EF Q unbuffered (griffin.yml Stream FIFO parts).

Sheet: /COMPOSITOR/	
File: compositor.kicad_sch	
Title:	
Size: USLetter	Date:
KiCad E.D.A. 9.0.7	Id: 16/21


+5V
U47
RGB_OUT11
RGB_OUT10
RGB_OUT9
RGB_OUT8
RGB_OUT7
RGB_OUT6
RGB_OUT5
RGB_OUT4
A0
A1
A2
A3
A4
A5
A6
A7
VCC
GND
G1
G2
Y0
Y1
Y2
Y3
Y4
Y5
Y6
Y7
VR3
VR2
VR1
VR0
VG3
VG2
VG1
VG0
74AC541
C47
100nF
U48
RGB_OUT3
RGB_OUT2
RGB_OUT1
RGB_OUT0
VGA_HSYNC
VGA_VSYNC
VB3
VB2
VB1
VB0
HS_BUF
VS_BUF
C48
R50
R51
R52
R53
R54
R55
R56
R57
R58
R59
R60
R61
R62
R63
R64
R65
R66
R67
R68
R69
R70
1k
3.9k
100R
VR1_MID
VG1_MID
VB1_MID
VGA_R
VGA_G
VGA_B
R71
R72
HS_OUT
VS_OUT
J22
SHIELD
SHIELD__1
DE-15 VGA
Ladders and shunts at the DE-15, not at the buffers; 12 DAC traces short
and matched (griffin.yml VGA entry). yml BOM says 21x 1k; this topology uses 15.
Sheet: /VGA_OUT/
File: vga.kicad_sch
Title:
Size: USLetter
Date:
Rev:
KiCad E.D.A. 9.0.7
Id: 17/21

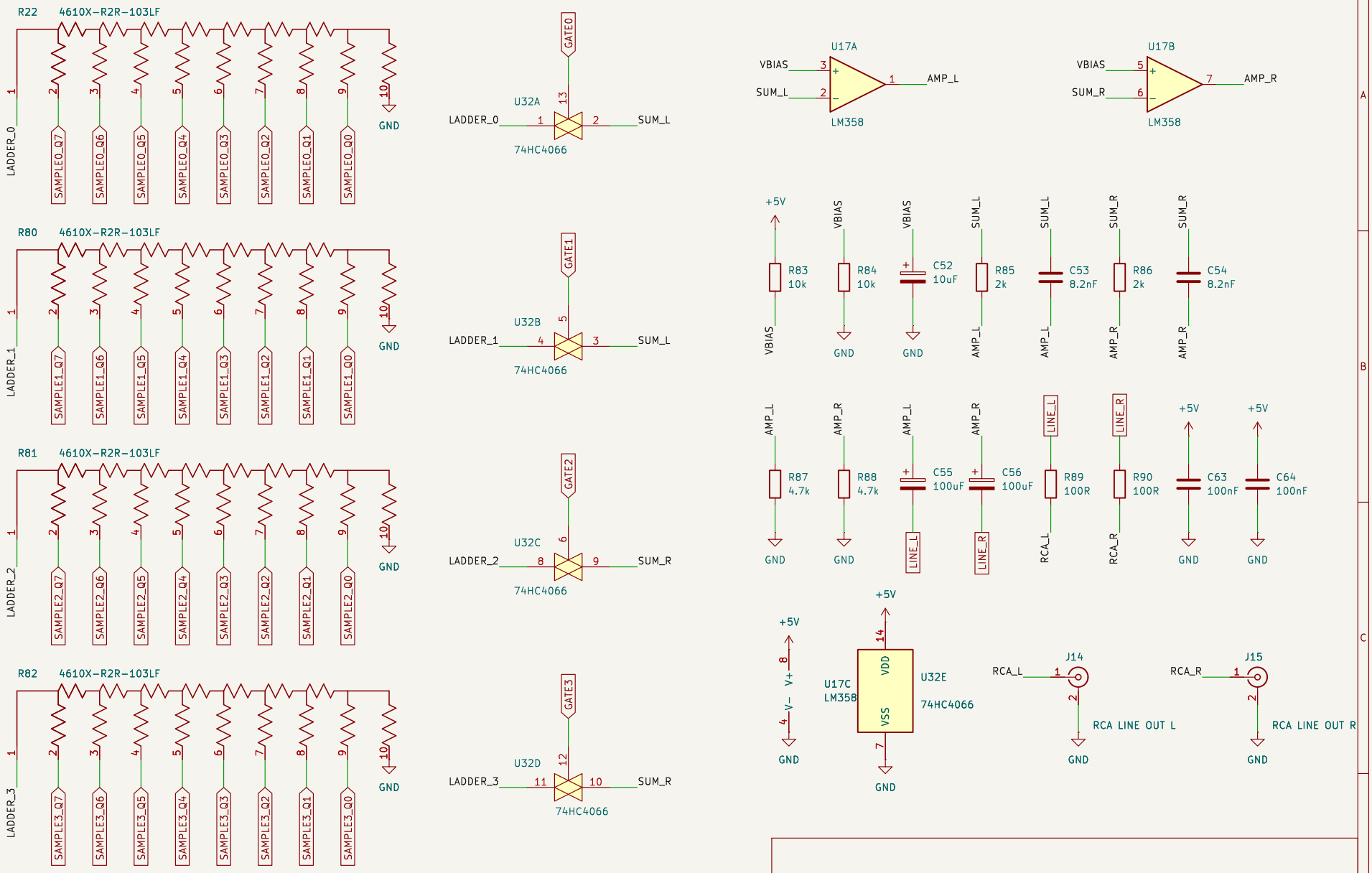

U45
74AHCT541
D0
D1
D2
D3
D4
D5
D6
D7
D_AUDIO_FIFO0
D_AUDIO_FIFO1
D_AUDIO_FIFO2
D_AUDIO_FIFO3
D_AUDIO_FIFO4
D_AUDIO_FIFO5
D_AUDIO_FIFO6
D_AUDIO_FIFO7
C50
100nF
U46
74AHCT541
D8
D9
D10
D11
D12
D13
D14
D15
D_AUDIO_FIFO8
D_AUDIO_FIFO9
D_AUDIO_FIFO10
D_AUDIO_FIFO11
D_AUDIO_FIFO12
D_AUDIO_FIFO13
D_AUDIO_FIFO14
D_AUDIO_FIFO15
C51
100nF
U28
AUDIO
GATE3
SAMPLE_R3
SAMPLE_EF2
SAMPLE_EF1
SAMPLE_EF0
SAMPLE_EF3
CONTROL_EF
HBLANK
GATE0
SAMPLE_R0
RESET
GATE1
CONTROL_Q2
SAMPLE_R1
CONTROL_Q1
CONTROL_Q0
GATE2
SAMPLE_R2
CONTROL_Q7
CONTROL_Q6
CONTROL_Q5
CONTROL_Q4
CONTROL_Q3
PIXEL_CLK
CONTROL_R
JTAG_TCK
PORTS_TO_AUDIO_TD
AUDIO_TO_ENGINE_TD
JTAG_TMS
C41
100nF
R157
0R
AUDCTL_FIFO_W
AUDCTL_FIFO_W_FIFO_SIDE
U29
IDT7200L15P
R39
0R
CONTROL_R
CONTROL_Q0
CONTROL_Q1
CONTROL_Q2
CONTROL_Q3
CONTROL_Q4
CONTROL_Q5
CONTROL_Q6
CONTROL_Q7
TP7
CONTROL_EF
C40
100nF
Audio FIFO write buffers U45, U46: AHCT, not HCT;
write-data path only (griffin.yml Stream FIFO parts).
The main bus does not run to the audio section.
Sheet: /AUDIO/
File: audio.kicad_sch
Title:
Size: USLetter
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Sheet: /AUDIO_FIFO/
File: audio-fifo.kicad_sch

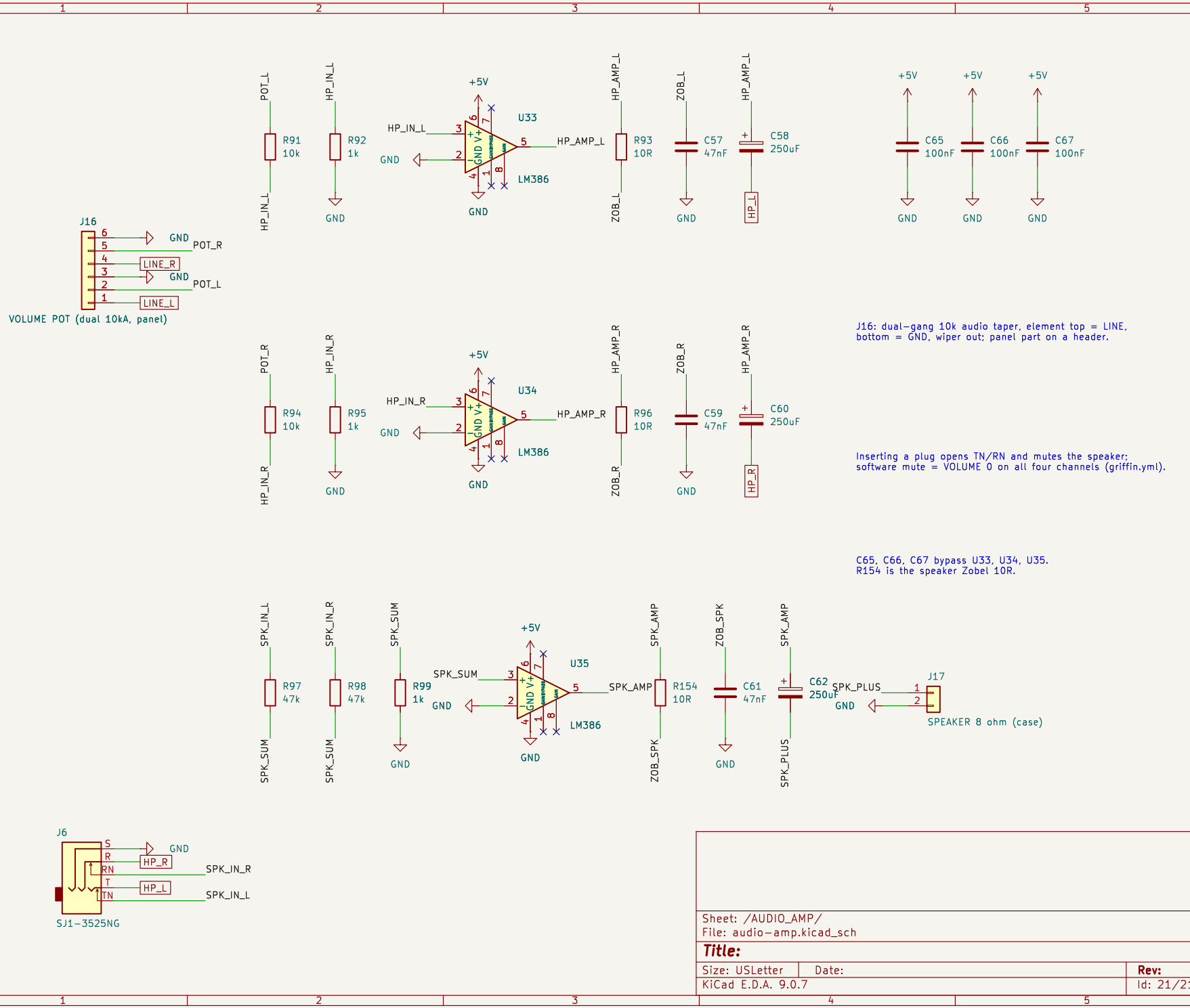
Title:

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Offset-binary samples: 0x80 = 2.5 V = silence; no inverter, no pulls on Q.
Summer biased at VBIAS 2.5 V, 2k || 8.2 nF (pole ~9.7 kHz), 4.7k class-A bias, 100 uF to LINE.
Line out fixed level ~0.88 Vrms. Not yet breadboarded (griffin.yml issues).
C63 bypasses U32, C64 bypasses U17.

Sheet: /AUDIO_DAC/	
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Title:	
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POT_L
HP_IN_L
R91
10k
R92
1k
GND
+5V
HP_IN_L
U33
GND V+
HP_AMP_L
LM386
HP_AMP_L
R93
10R
ZOB_L
C57
47nF
C58
250uF
HP_L
+5V
C65
100nF
C66
100nF
C67
100nF
J16
GND
POT_R
LINE_R
GND
POT_L
LINE_L
VOLUME POT (dual 10kA, panel)
POT_R
HP_IN_R
R94
10k
R95
1k
HP_IN_R
U34
HP_AMP_R
R96
10R
ZOB_R
C59
47nF
C60
250uF
HP_R
J16: dual-gang 10k audio taper, element top = LINE, bottom = GND, wiper out; panel part on a header.
Inserting a plug opens TN/RN and mutes the speaker; software mute = VOLUME 0 on all four channels (griffin.yml).
C65, C66, C67 bypass U33, U34, U35.
R154 is the speaker Zobel 10R.
SPK_IN_L
SPK_IN_R
SPK_SUM
R97
47k
R98
47k
R99
1k
U35
SPK_AMP
R154
10R
ZOB_SPK
C61
47nF
C62
250uF
SPK_PLUS
J17
SPEAKER 8 ohm (case)
J6
S
R
RN
T
TN
HP_R
HP_L
SPK_IN_R
SPK_IN_L
SJ1-3525NG
Sheet: /AUDIO_AMP/
File: audio-amp.kicad_sch
Title:
Size: USLetter
Date:
Rev:
KiCad E.D.A. 9.0.7
Id: 21/21